Ben Limpanukorn

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Education

2022 – Present **Ph.D. Candidate, University of California, Los Angeles (UCLA)** in Computer Science.

2021 – 2022 **M.S., University of California, Los Angeles (UCLA)** in Computer Science.
Capstone: *Challenges and Methods for Testing NVMe SSD Devices using Software Fuzzing*.

2017 – 2021 **B.S., University of California, Los Angeles (UCLA)** in Computer Science and Engineering.

Fellowships

2025 – 2026 **Amazon Trainium Fellowship**, Amazon Web Services.
Research on testing and optimizing ML compilers and kernels for AWS Trainium.

Publications

1. **Ben Limpanukorn**, Jiyuan Wang, Hong Jin Kang, Zitong Zhou, and Miryung Kim. "Fuzzing MLIR Compilers with Custom Mutation Synthesis". In: *2025 IEEE/ACM 47th International Conference on Software Engineering (ICSE)*. 2025, pp. 217–229. DOI: 10.1109/ICSE55347.2025.00037.

2. Jai Parera, Nathan Huey, **Ben Limpanukorn**, and Miryung Kim. "Chrysalis: A Lightweight Logging and Replay Framework for Metamorphic Testing in Python". In: *Proc. 40th IEEE/ACM International Conference on Automated Software Engineering (ASE), Tool Demonstration Track*. 2025, pp. 4045–4048. DOI: 10.1109/ASE63991.2025.00371.

3. Jiyuan Wang, Yuxin Qiu, **Ben Limpanukorn**, Hong Jin Kang, Qian Zhang, and Miryung Kim. "DuoReduce: Bug Isolation for Multi-layer Extensible Compilation". In: *Proc. ACM Softw. Eng.* 2.FSE (June 2025). DOI: 10.1145/3715747.

4. Joshua M. Clifford, **Ben Limpanukorn**, and Edward S. Jimenez. "An Improved Process to Colorize Visualizations of Noisy X-Ray Hyperspectral Computed Tomography Scans of Similar Materials". In: *2022 IEEE Nuclear Science Symposium and Medical Imaging Conference (NSS/MIC)*. 2022, pp. 1–4. DOI: 10.1109/NSS/MIC44845.2022.10399061.

5. Joshua Clifford, Emily Kemp, **Ben Limpanukorn**, and Edward S. Jimenez. "A Process to Colorize and Assess Visualizations of Noisy X-Ray Computed Tomography Hyperspectral Data of Materials with Similar Spectral Signatures". In: *2021 IEEE Nuclear Science Symposium and Medical Imaging Conference (NSS/MIC)*. 2021, pp. 1–8. DOI: 10.1109/NSS/MIC44867.2021.9875740.

Papers Under Submission

6. **Ben Limpanukorn**, Andrew Hu, Rajdeep Mondal, Adam Thieme, Jared Velasquez, Brian Yang, Yutong Zhou, and Miryung Kim. "GramCon: Stateful, Conditioned Grammar-Based Generation with Goal-Directed Backtracking". Under submission. 2026.

7. **Ben Limpanukorn**, Yuxin Qiu, Jiyuan Wang, Qian Zhang, Tony Nowatzki, and Miryung Kim. "Parametric Performance Models for Selecting and Testing Kernel Optimizations". Under submission. 2026.

8. **Ben Limpanukorn**, Jiyuan Wang, Seongmin Lee, Yuxin Qiu, Qian Zhang, and Miryung Kim. "Let Error Messages Teach the Fuzzer: Rule-based Repair for Bespoke Fuzzing with LLMs". Under submission. 2026.

9 Yuxin Qiu, **Ben Limpanukorn**, Seongmin Lee, Jiyuan Wang, Qian Zhang, and Miryung Kim. "Tensor Algebraic Property Skeletons: Amplifying Property-Based Testing for AI Compilers". Under submission. 2026.

Preprints

10 **Ben Limpanukorn**, Yanjun Wang, Zach Patterson, Pranav Garg, Murali Krishna Ramanathan, Xiaofei Ma, Anoop Deoras, and Miryung Kim. *Structural Code Search using Natural Language Queries*. 2025. arXiv: 2507.02107 [cs.SE]. URL: https://arxiv.org/abs/2507.02107.

Talks and Posters

2026 — ***Let Error Messages Teach the Fuzzer: Rule-based Repair for Bespoke Fuzzing with LLMs***, research talk, Southern California Programming Languages and Systems Workshop (SoCal PLS).

— ***Parametric Performance Models for Selecting and Testing Kernel Optimizations***, poster, Amazon Research Day.

2025 — ***Chrysalis: A Lightweight Logging and Replay Framework for Metamorphic Testing in Python***, tool demonstration and poster, ASE 2025 Tool Demonstration Track.

— ***Triton to NKI Kernel Transpilation***, poster, UCLA–Amazon Science Hub Fall Showcase.

Experience

2022 – Present — **Graduate Student Researcher** SEAL Lab at UCLA.

- Designed SynthFuzz, a fuzzer for MLIR compilers that automatically infers context-sensitive mutations from test cases, matching the branch coverage of domain-specific fuzzers with minimal manual effort [1].
- Used LLM-inferred repair rules to increase deepest-stage validation by 10.3× across four MLIR dialects.
- Contributed to DuoReduce, an MLIR delta debugger that jointly reduces IR code and compilation passes, cutting pass-search time from up to 145 hours to 9.5 minutes [3].
- Guided LLM-based kernel optimization with calibrated performance models, achieving a 2.03× geomean speedup across 36 AWS Trainium (NKI) kernels.
- Developed workflows that translate natural-language software specifications into executable property-based tests for MLIR compiler passes.
- Advised graduate students whose metamorphic-testing framework was accepted to the ASE 2025 Tool Demonstration Track [2].

Jun – Dec 2024 — **Applied Science Intern** Amazon Web Services.

- Developed a new approach to structural code search by translating natural-language queries into domain-specific languages such as Semgrep and GQL using large language models.
- Created a new benchmark for structural code search, consisting of 400 queries over 10 Java projects, to evaluate the performance of such tools.
- Outperformed embedding-based search by up to 57 percentage points in F1 score.

Experience (continued)

2020 – 2024 **R&D Graduate Student Intern** Sandia National Laboratories.
- Productionized object detection models for next-generation millimeter-wave scanners, resulting in a 6× reduction in inference time and improved precision/recall compared to a baseline model.
- Developed new methods to colorize/visualize hyperspectral CT data by combining dimension reduction (e.g. UMAP and T-SNE) with generalized linear models to improve the interpretability of spectrally similar materials.
- Used Vitis HLS to develop an FPGA implementation of a List-Mode MLEM reconstruction algorithm, achieving a 5× speedup compared to an equivalent CPU implementation.

2019 – 2020 **DevOps Engineer** Camino Financial
- Built CI/CD and ChatOps systems that increased the deployment rate by 25%, enabled zero-downtime deployments, and improved production monitoring.
- Streamlined development workflows to reduce unnecessary operational overhead.

2018 – 2019 **Research Assistant** LEMUR Lab at UCLA
- Extended the CoLo cooperative-localization simulator to model signal strength and communication cost, and automated data collection for physical experiments.

Teaching

Fall 2024 **Teaching Assistant, CS 230: Software Engineering**, UCLA.
Led office hours; developed practice and final-exam questions; and graded exams, reports, and capstone projects.

Skills

Languages	Python, Rust, C++, C, Go, Bash, Fish, LaTeX, Lean 4
Libraries/Frameworks	PyTorch, TensorFlow, NumPy, pandas, Matplotlib
Compilers	MLIR (dialects, passes, xDSL), LLVM, CIRCT, AWS Neuron/NKI
Systems/Testing	ANTLR, AFL++, LibAFL, Docker